🔬 General Science — Flowchart Memory Map

Physics · Chemistry · Biology — Motion, Sound, Light, Electricity, Acids & Bases, Metals, Human Body, Vitamins & Diseases — drawn as trees, tables & mnemonics so it sticks in one glance.

Target post: ASO · Tier-I General Awareness · NCERT-level scope. Every block tagged by priority.

P1 High-yield — do first P2 Medium P3 Low / rare 🧠 = mnemonic

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🗓️ **2026-07-04** · **Concept of the day:** Acids & Bases — pH scale, indicators rotates daily

🧭 Science in 3 Branches P1

Start here. Every question below hangs off one of these three trunks — Physics (energy & forces), Chemistry (matter & reactions), Biology (life).

GENERAL SCIENCE

- ⚙️ **Physics** → Motion & Force · Sound · Light · Electricity · Heat & Energy
- 🧪 **Chemistry** → Acids · Bases · Salts · Metals & Non-metals · Periodic Table · Everyday Chemistry
- 🌿 **Biology** → Human Body Systems · Vitamins & Nutrition · Diseases · Cell & Genetics

🧠 **"PCB"** = **P**hysics → **C**hemistry → **B**iology. In SSC, Biology & everyday-Chemistry facts repeat most; pure numerical Physics is rarer but easy marks.

🚴 Motion & Force P1

Newton's laws + SI units are near-certain 1-markers. Learn the units table cold.

Newton's 3 Laws of Motion

LAW	STATEMENT	ALSO CALLED / KEY IDEA	TAG
1st Law	A body stays at rest or in uniform motion unless acted on by an external force	Law of **Inertia**; defines force	P1

LAW	STATEMENT	ALSO CALLED / KEY IDEA	TAG
2nd Law	Force = rate of change of momentum → **F = m·a**	Gives the formula for force (unit: newton)	P1
3rd Law	Every action has an **equal & opposite** reaction	Rocket propulsion, recoil of gun, swimming	P1

🧠 **"Inertia → Formula → Reaction"** for laws 1-2-3. Inertia ∝ mass (heavier = more inertia).

SI Base Units (the "must-know 7")

QUANTITY	SI UNIT	SYMBOL
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

🧠 **"Many Kids Study At King's College Daily"** → Metre, Kilogram, Second, Ampere, Kelvin, Candela ... (Mole in the mix). Force = **newton (N)**, Energy = **joule (J)**, Power = **watt (W)**, Pressure = **pascal (Pa)** — these are *derived*, not base.

Speed · Velocity · Acceleration · Momentum

QUANTITY	FORMULA	SCALAR / VECTOR	SI UNIT
Speed	distance ÷ time	Scalar	m/s
Velocity	displacement ÷ time	Vector (has direction)	m/s
Acceleration	change in velocity ÷ time	Vector	m/s^2
Momentum	p = mass × velocity	Vector	kg·m/s

Scalar vs Vector: Scalar = magnitude only (speed, distance, mass, energy). Vector = magnitude + direction (velocity, displacement, force, momentum, acceleration).

🔊 Sound P1

SOUND — a longitudinal, mechanical wave

Needs medium Cannot travel in **vacuum** (unlike light)

Speed order **Solid > Liquid > Gas** — fastest in solids (steel), slowest in gases

In air ≈ **343 m/s** at ~20 °C (rises with temperature)

RANGE	FREQUENCY	NOTE	TAG
Infrasonic	below 20 Hz	Heard by whales, elephants; not humans	P2
Audible (human)	**20 Hz – 20,000 Hz**	Normal hearing range	P1
Ultrasonic	above 20,000 Hz	Bats, dolphins; used in SONAR, ultrasound scan, cleaning	P1

Echo: reflected sound heard distinctly — needs a gap of at least **0.1 s** ⇒ reflector must be ≥ ~17 m away. **SONAR** (SOund Navigation And Ranging) uses ultrasonic waves to measure ocean depth & detect objects underwater.

🧠 Sound is **SLOW** vs light: it needs a medium and moves fastest in **Solids** — **"S for Solid = Speedy sound."** Pitch ∝ frequency; loudness ∝ amplitude.

💡 Light P1

LIGHT — travels in straight lines; speed ≈ 3×10^8 m/s in vacuum

REFLECTION (mirrors)

Laws ∠incidence = ∠reflection; both & normal lie in one plane

Concave Converging — real/virtual image; used in torches, shaving mirror

REFRACTION (lenses)

Cause Bending of light as it changes medium (speed change)

Convex lens Converging — real/virtual; corrects hypermetropia

Convex	Diverging — always virtual, erect, small; vehicle rear-view	**Concave lens**	Diverging — always virtual, erect; corrects myopia

Dispersion: white light splits into 7 colours through a prism — **VIBGYOR** (Violet, Indigo, Blue, Green, Yellow, Orange, Red). **Red** bends least (longest λ), **Violet** bends most. Rainbow = dispersion + total internal reflection in raindrops.

Total Internal Reflection (TIR): light in a denser medium hitting the boundary beyond the **critical angle** reflects fully. Basis of **optical fibres**, diamond sparkle, mirage.

Human Eye — Defects & Correction (HIGH-yield)

DEFECT	ALSO CALLED	PROBLEM	CORRECTING LENS	TAG
Myopia	Short-sightedness	Can't see **distant** objects; image forms *before* retina	**Concave** (diverging)	P1
Hypermetropia	Long-sightedness	Can't see **near** objects; image forms *behind* retina	**Convex** (converging)	P1
Presbyopia	Age-related	Loss of accommodation with age (both near & far)	Bifocal lens	P2
Astigmatism	—	Uneven cornea curvature; blurred vision	Cylindrical lens	P3

🧠 **Myopia → Minus (concave); Hyper → plus (convex).** "Near-sighted needs Concave, Far-sighted needs Convex." Retina has rods (dim light) & cones (colour).

⚡ Electricity P1

OHM'S LAW — V = I × R

V (Voltage)	Potential difference — unit **volt (V)**; measured by voltmeter (in parallel)
I (Current)	Flow of charge — unit **ampere (A)**; measured by ammeter (in series)
R (Resistance)	Opposition to current — unit **ohm (Ω)**
Power	**$P = V \times I = I^2R$** — unit **watt (W)**; 1 kWh = 1 "unit" of electricity

FEATURE	SERIES CIRCUIT	PARALLEL CIRCUIT
Current	Same through all components	Divides among branches
Voltage	Divides across components	Same across all branches
Total resistance	Adds up ($R_1+R_2+...$) — increases	Less than smallest ($1/R = \Sigma 1/R_i$)
If one fails	Whole circuit breaks (old fairy lights)	Others keep working (house wiring)

Fuse: a safety device — a thin wire of **low melting point & high resistance** connected in **series**; melts & breaks the circuit on excess current (short circuit / overload). **Conductors** = metals (Cu, Ag, Al — silver best); **Insulators** = rubber, glass, plastic, wood.

🧠 **Home wiring = PARALLEL** (so each appliance gets full 230 V and one failure doesn't kill the rest). **V-I-R**: cover the letter you want in the triangle V / (I·R).

🧪 Acids, Bases & Salts P1

The pH scale + everyday acid names are asked almost every shift.

pH Scale (0 – 14)



Acid < 7 < Base (alkali). pH 7 = neutral (pure water). Lower pH = more acidic (more H^+ ions).

Indicators

INDICATOR	IN ACID	IN BASE
Litmus	Red	Blue
Phenolphthalein	Colourless	Pink
Methyl orange	Red / Pink	Yellow
Turmeric	Stays yellow	Red / brown

🧠 **Acid turns blue litmus RED** ("Acid → Red"). Phenolphthalein is **Pink in base** only.

COMMON ACIDS

Lab acids

- HCl (hydrochloric)
- H_2SO_4 (sulphuric)
- HNO_3 (nitric)

Everyday

- Vinegar = Acetic
- Curd = Lactic
- Lemon = Citric
- Ant sting = Formic

COMMON BASES

Strong

- NaOH (caustic soda)
- KOH (caustic potash)

Everyday

- Soap
- Milk of magnesia (antacid)
- Baking soda ($NaHCO_3$)
- Lime water $Ca(OH)_2$

Neutralization: Acid + Base → **Salt + Water** (e.g. HCl + NaOH → NaCl + H_2O). Antacids (mild base) neutralise excess stomach acid (HCl). **Everyday acids:** Vinegar→Acetic · Curd/Milk→Lactic · Orange/Lemon→Citric · Tamarind→Tartaric · Tomato→Oxalic · Ant/Nettle→Formic.

Metals & Non-metals P1

Reactivity Series (most → least reactive)

HIGH reactivity →→→ **LOW reactivity**

K	Na	Ca	Mg	Al	Zn	Fe	Pb	(H)	Cu	Hg	Ag	Au

🧠 **"Please Send Charlie's Monkeys And Zebras In Lead Cages Securely Guarded"** ... or the classic **K Na Ca Mg Al Zn Fe Pb H Cu Hg Ag Au.** Metals above H displace hydrogen from acids. **Au (gold) & Ag (silver)** = least reactive → found free in nature.

Metals vs Non-metals

PROPERTY	METALS	NON-METALS
State	Solid (except **Mercury** — liquid)	Solid, liquid (Br) or gas

PROPERTY	METALS	NON-METALS
Lustre	Shiny	Dull (except iodine, diamond)
Conductivity	Good conductor (heat & electricity)	Poor (except **graphite** — conducts)
Malleable / Ductile	Yes (Au most malleable & ductile)	No — brittle
Oxide nature	Basic	Acidic
Nature of ion	Lose electrons → cation (+)	Gain electrons → anion (−)

Alloys (composition)

ALLOY	COMPOSITION	USE	TAG
Brass	Copper + Zinc	Utensils, decorative items	P1
Bronze	Copper + Tin	Statues, medals, coins	P1
Steel	Iron + Carbon	Construction, tools	P1
Stainless steel	Iron + Chromium + Nickel + Carbon	Cutlery, rust-free items	P2
Solder	Lead + Tin	Joining/welding electrical wires	P2
Amalgam	Mercury + another metal	Dental fillings	P3

Corrosion / Rusting: Iron + Oxygen + Water (moisture) → **hydrated iron oxide (rust, reddish-brown)**. Prevented by galvanising (zinc coating), painting, oiling, chrome plating. Aluminium forms a protective oxide layer (doesn't rust away). Copper corrodes to green (basic copper carbonate).

🧠 **Brass = Zinc, Bronze = Tin.** "BraZZ has Zinc." Both start with Copper. Solder = Lead + Tin (joins wires).

🫁 Human Body Systems P2

🍽️ Digestive

Mouth → oesophagus → stomach (HCl) → small intestine (absorption) → large

❤️ Circulatory

Heart has **4 chambers** (2 atria + 2 ventricles); pumps blood via arteries/veins P1

RBC carry O_2 · WBC fight infection

🫁 Respiratory

Nose → trachea → lungs → alveoli (gas exchange). Diaphragm aids breathing P2

intestine P2

Largest gland: Liver

🧠 Nervous

Brain + spinal cord + nerves. Neuron = functional unit; controls all responses P2

💧 Excretory

Kidneys filter blood → urine via ureter → bladder. Nephron = functional unit P3

🦴 Skeletal

206 bones in adult. Longest: Femur · Smallest: Stapes (ear) · Strongest: Jaw/femur P2

"LARGEST / SMALLEST / ..."	ANSWER	TAG
Largest organ of body	**Skin**	P1
Largest internal organ / gland	**Liver** (largest gland)	P1
Longest bone	**Femur** (thigh)	P1
Smallest bone	**Stapes** (in the ear)	P1
Universal blood **donor**	**O-negative**	P1
Universal blood **recipient**	**AB-positive**	P1
Largest / smallest cell	Nerve cell (longest) / — ; RBC has no nucleus	P2
Number of chambers in heart	**4** (2 atria, 2 ventricles)	P1

🧠 **Skin = biggest organ, Liver = biggest gland, Femur = longest bone, Stapes = smallest bone.** Blood: **O^- gives to all (donor), AB^+ takes from all (recipient).**

🥦 Vitamins & Deficiency Diseases P1

One of the highest-frequency Science topics — learn chemical name + deficiency together.

VITAMIN	CHEMICAL NAME	DEFICIENCY DISEASE	RICH SOURCE
A	Retinol	Night blindness, xerophthalmia	Carrot, papaya, milk
B1	Thiamine	Beriberi	Whole grains, pulses
B2	Riboflavin	Cracked lips / cheilosis	Milk, egg, green veg
B3	Niacin	Pellagra	Meat, groundnut

VITAMIN	CHEMICAL NAME	DEFICIENCY DISEASE	RICH SOURCE
B12	Cyanocobalamin	Anaemia (pernicious)	Meat, liver, egg
C	Ascorbic acid	Scurvy (bleeding gums)	Amla, citrus fruits
D	Calciferol	Rickets (children), osteomalacia (adults)	Sunlight, fish, egg yolk
E	Tocopherol	Reproductive weakness / infertility	Vegetable oils, nuts
K	Phylloquinone	Poor blood clotting (excess bleeding)	Green leafy vegetables

🧠 **"A = eyes (night blindness), C = Scurvy, D = Rickets/bones, K = Klotting."** Fat-soluble vitamins = **A, D, E, K**; rest (B, C) are water-soluble.

Vitamin D is the "sunshine vitamin" (made in skin from sunlight). **Vitamin C** is destroyed by cooking/heat. Vitamin B & C dissolve in water, so excess is flushed out.

🦠 Diseases — by Causative Agent P1

CAUSED BY	DISEASES	TAG
Bacteria	Tuberculosis (TB), Typhoid, Cholera, Tetanus, Diphtheria, Plague, Leprosy	P1
Virus	Polio, Dengue, Rabies, Influenza, Chickenpox, Measles, Hepatitis, COVID-19, AIDS	P1
Protozoa	Malaria (Plasmodium), Amoebiasis, Kala-azar, Sleeping sickness	P1
Fungus	Ringworm, Athlete's foot, Dandruff	P2
Worm	Filariasis (elephantiasis), Tapeworm, Ascariasis	P3

Vector-borne (who carries it)

VECTOR	DISEASE(S)	NOTE
Female Anopheles mosquito	Malaria	Protozoan (Plasmodium)
Aedes mosquito	Dengue, Chikungunya, Zika, Yellow fever	Viral; bites in daytime
Culex mosquito	Filariasis, Japanese encephalitis	—
Housefly	Cholera, Typhoid, Dysentery	Contaminates food

VECTOR	DISEASE(S)	NOTE
Sandfly	Kala-azar (Leishmaniasis)	Protozoan

🧠 **Anopheles = Malaria; Aedes = Dengue/Chikungunya/Zika; Culex = Filariasis.** "A-A-C" for the 3 mosquitoes. Malaria is **protozoan**, Dengue is **viral** — classic trap.

Deficiency (non-communicable) diseases aren't caused by germs — e.g. Scurvy, Rickets, Goitre (iodine), Anaemia (iron). **Goitre** = enlarged thyroid from iodine deficiency.

⚡ 60-Second Rapid Revision P1

PROMPT	ANSWER
SI unit of force / energy / power	Newton / Joule / Watt
Newton's 1st law is also called	Law of Inertia
Speed of sound in air	≈ 343 m/s (fastest in solids)
Human audible range	20 Hz – 20,000 Hz
SONAR uses which waves	Ultrasonic (>20 kHz)
Myopia corrected by	Concave (diverging) lens
Hypermetropia corrected by	Convex (converging) lens
7 colours of light	VIBGYOR
Ohm's law	V = I × R (Power P = VI)
House wiring is in	Parallel
Fuse is made of	Low-melting, high-resistance wire (in series)
Neutral pH / acid / base	7 / below 7 / above 7
Acid turns blue litmus	Red
Curd / Vinegar / Lemon acid	Lactic / Acetic / Citric
Brass / Bronze composition	Cu+Zn / Cu+Sn

PROMPT	ANSWER
Steel composition	Iron + Carbon
Least reactive (found free) metals	Gold & Silver
Rusting needs	Iron + Oxygen + Water
Largest organ / gland	Skin / Liver
Longest / smallest bone	Femur / Stapes
Chambers of human heart	4
Universal donor / recipient	O-negative / AB-positive
Vitamin C name / deficiency	Ascorbic acid / Scurvy
Vitamin D name / deficiency	Calciferol / Rickets
Vitamin A deficiency	Night blindness
Vitamin K function	Blood clotting
Beriberi caused by lack of	Vitamin B1 (Thiamine)
Malaria is caused by	Protozoan (Plasmodium), via Anopheles
Dengue / Chikungunya vector	Aedes mosquito (viral)
TB / Typhoid / Cholera type	Bacterial
Polio / Rabies / Dengue type	Viral
Goitre caused by lack of	Iodine

PYQ pattern: "Which vitamin prevents scurvy?" → Vitamin C. · "Myopia is corrected by which lens?" → Concave. · "SI unit of electric current?" → Ampere. · "Malaria is caused by?" → Plasmodium (protozoan).